

REED PUTNAM PRACTICE · FALL 2026

Bijections, Double Counting, Recursions & Generating Functions

Week 5 · the shape of the answer picks the tool

Thu Oct 29 / Fri Oct 30

The toolbox (1)

Bijection

$|A| = |B|$ once you write down a map $A \rightarrow B$ and its inverse. To count A , find a B you already know: k -subsets $\binom{n}{k}$; E/N paths to (a, b) : $\binom{a+b}{a}$; compositions of n into k parts: $\binom{n-1}{k-1}$; stars and bars: $\binom{n+k-1}{k}$.

- Double counting: for $I \subseteq A \times B$, row sums = column sums of the incidence matrix.
- Handshake: $\sum_v \deg v = 2|E|$; “count the pairs (element, set containing it).”
- The payoff is often an inequality, after bounding one side.

The toolbox (2)

- > **Recursion:** condition on one feature (is n in the set? the last step? the largest element?), e.g. $D_n = (n-1)(D_{n-1} + D_{n-2})$.
- > Record the **base cases and the range of n** where the recurrence holds.
- > $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$ has solutions r^n , r a root of $x^k - c_1 x^{k-1} - \dots - c_k$.
- > **Generating functions:** $A(x) = \sum a_n x^n$, a formal series; convergence is irrelevant.
- > **Independent choices multiply:** $c_n = \sum_k a_k b_{n-k}$ means $C = AB$.
- > $\frac{1}{1-x} = \sum x^n$, $\frac{1}{(1-x)^k} = \sum_n \binom{n+k-1}{k-1} x^n$, $\frac{1}{1-x-x^2} = \sum F_{n+1} x^n$.

Example 1

Double counting: committees with a chair

$$\sum_{k=0}^n k \binom{n}{k} = n 2^{n-1}.$$

→ work it on the board (~2 min)

Example 2

Bijection: the reflection principle

The number of E/N lattice paths from $(0, 0)$ to (n, n) that never rise above the diagonal $y = x$ is

$$\binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1} \binom{2n}{n}.$$

→ work it on the board (~4 min)

Example 3

Generating functions: Euler's partition identity

The number of partitions of n into **distinct** parts equals the number of partitions of n into **odd** parts.

→ work it on the board (~3 min)

How to recognize it

- > “Find the number of...”: the **shape of the answer** picks the tool.
- > Closed form $\binom{a}{b}$ or 2^{n-2} : **bijection** with subsets or paths.
- > Fibonacci-like sequences: **recursion** on the largest element or the last step.
- > An inequality between counts: **double counting** plus one estimate.
- > Polynomial in n , symmetric, sum of products: **generating function**.

Pitfalls

- > “Every X produces a Y ” is a map, not a bijection: write the inverse.
- > Divide by $k!$ only when all rearrangements are distinct; check the small cases.

Now you

Problems 1–4.

1–2 warm-ups · 3–4 easy Putnam · 5–6 take-home

35 minutes, groups of 2–3

Solutions posted after Friday.